Program: **Diploma in Renewable Energy**	
Course Code: **3292**	Course Title: **Electrical & Renewable Energy Measuring Systems**
Semester: **3**	Credits: **3**
Course Category: **Program Core**	
Periods per week: **3 (L:2, T:1, P:0)**	Periods per semester: **45**

Course Objectives:

- To impart basic knowledge of electrical measurements & instrumentation.
- To understand the fundamental concepts of different measuring instruments.
- To familiarize measurement of various renewable energy parameters.

Course Prerequisites:

Topic	Course code	Course Name	Semester
Basic concepts of Electrical & Electronics Engineering		Fundamentals of Electrical & Electronics Engineering	2
Basics concepts of different renewable sources		Basic Electrical Engineering & Renewable Technology.	2

Course Outcomes:

On completion of the course, the students will be able to:-

COn	Description	Duration (Hours)	Cognitive Level
CO1	Identify the operation of different electrical measuring instruments.	10	Applying
CO2	Summarize the measuring methods of various circuit parameters and physical quantities.	12	Understanding

CO3	Identify the measuring methods of electrical power and energy.	11	Applying
CO4	Summarize the principle of different solar and wind energy measuring instruments.	10	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

On completion of the course, the students will be able to:-

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Identify the operation of different electrical measuring instruments.**		
M1.01	Classify different electrical measuring instruments	2	Understanding
M1.02	Explain the production of various torques in electrical measuring instruments	2	Understanding
M1.03	Illustrate the working of PMMC & MI instruments	3	Understanding
M1.04	Solve simple problems to extent the range of PMMC instruments using shunts and multipliers	3	Applying
Contents: **Basics of measurement**- Definition- standards- errors- need for calibration. **Classification of instruments**- Absolute & secondary- Indicating, Recording & Integrating- Analog & digital- Null & deflection type.			

Production of torque- deflection, damping & controlling torque (definition and methods of production only) - pointers- scales- supporting systems. Construction and working- PMMC & MI instruments- errors- comparison. **Range extension of meters**- shunts & multipliers- simple problems.			
CO2	**Summarize the measuring methods of various circuit parameters and physical quantities.**		
M2.01	Classify different measuring methods of resistance.	3	Understanding
M2.02	Illustrate the measurement of inductance and capacitance using bridge circuits.	3	Understanding
M2.03	Summarize the concepts of transducers and sensors.	3	Understanding
M2.04	Illustrate the operation of CRO and DSO	3	Understanding
	Series Test – I	1	
Classification and measurement of Resistance- Ammeter Voltmeter method- Kelvin's double bridge- Wheatstone's bridge(Derivation and circuit needed)- Megger- Earth resistance measurement- Multimeters. **Measurement of inductance**- Maxwell's inductance bridge (Derivation and circuit needed) - Anderson's bridge (Derivation not needed). **Measurement of capacitance**- Schering's bridge (Derivation not needed). **Transducers and sensors** - Definition- applications- Measurement using transducers- principle of operation- Thermocouple- Thermistors- Strain gauges-Light sensors(Photoresistor, Photodiodes)-Ultrasonic Sensors-semiconductor sensors(basic concept only). **CRO**- working and block diagram- applications- measurement of voltage, frequency& phase difference- time base and synchronization- dual trace CRO. **Digital Storage Oscilloscope**- block diagram			
CO3	**Identify the measuring methods of electrical power and energy.**		
M3.01	Identify the principle of power measurement using dynamometer type wattmeter.	3	Applying
M3.02	Explain the construction and operation of induction type energy meter.	2	Understanding

<table>
<tr><td>M3.03</td><td>Summarize calibration methods of wattmeter and energy meter.</td><td>3</td><td>Understanding</td></tr>
<tr><td>M3.04</td><td>Explain the working of special-purpose measuring instruments.</td><td>3</td><td>Understanding</td></tr>
<tr><td colspan="4">Contents :

Measurement of power- Dynamometer type Wattmeter - construction- principle of operation- errors & compensation- Measurement of three phase power using two wattmeter method- simple mathematical problems using two wattmeter method
Measurement of energy- Induction type energy meter- construction- principle of operation- error & compensation.
Calibration of meters- Wattmeter- Energy meter- direct & phantom loading.

Special measuring instruments- single phase power factor meter- Frequency meter(reed & electro resonance types)- Instrument transformers (CT & PT)- Basic operation with diagrams</td></tr>
<tr><td>CO4</td><td colspan="3">Summarize the principle of different solar and wind energy measuring instruments</td></tr>
<tr><td>M4.01</td><td>Outline various terms related to solar energy.</td><td>3</td><td>Understanding</td></tr>
<tr><td>M4.02</td><td>Illustrate different solar radiation measuring instruments.</td><td>3</td><td>Understanding</td></tr>
<tr><td>M4.03</td><td>Summarize basic parameters related to wind energy measurement.</td><td>2</td><td>Understanding</td></tr>
<tr><td>M4.04</td><td>Explain the principle of wind energy measuring instruments.</td><td>2</td><td>Understanding</td></tr>
<tr><td></td><td>Series Test – II</td><td>1</td><td></td></tr>
<tr><td colspan="4">Contents :

Solar energy- important terms and definitions - Solar radiation-Terrestrial and extraterrestrial solar radiation-Beam radiation -Diffuse radiation-solar insolation-Solar irradiation-Air mass-AM0-AM1-AM1.5-AM2 radiation -solar constant

Solar radiation geometry-Latitude angle-declination angle-hour angle-solar altitude-zenith angle-solar azimuth angle-surface azimuth angle-tilt angle-incident angle(definition only)

Solar energy measuring instruments- Construction & working - pyrheliometer, pyranometer,- -sunshine recorder- Basic concepts of Pyrgeometer-Solarimeter -solar reference cell</td></tr>
</table>

Wind energy measurement- basic parameters- wind speed- vertical wind speed.- Beaufort wind scale-Effect of various parameters on wind speed- temperature- solar radiation- barometric pressure. **Wind energy measuring instruments**- - cup anemometer- ultrasonic anemometer (construction and working)-LIDAR for wind measurements (elementary concept only) Wind energy collectors- factors affecting the distribution of wind energy on the surface of earth.

Text / Reference

T/R	Book Title/Author
T1	A Course in Elec. & Electronics Measurements & Instrumentation: A K. Sawhney
T2	Non-Conventional Energy sources- G D Rai-Khanna Publishers
R1	Modern Electronic Instrumentation and Measurement Techniques: Helfrick & Cooper
R2	Electrical Measurement and Measuring Instruments - Golding & Waddis
R3	D.V.S.Murty, “Transducers and Instrumentation”, Second Edition, Prentice-Hall of India Private Limited, 2008
R4	ArunK.Ghosh, “Introduction to Measurements and Instrumentation”, Third Edition, PHI Learning Private Limited
R5	Non-conventional Energy Resources -B H Khan Mc Graw Hill Education
R6	D.P. Kothari, K. C. Singal, Rakesh Ranjan, “Renewable Energy Sources and Emerging Technologies” Prentice Hall India Learning Private Limited
R7	John Twidell and Tony Weir “Renewable Energy Resources”,Third Edition,Special Indian Edition.

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/108/105/108105153
2	https://nptel.ac.in/courses/112/104/112104300
3	https://nptel.ac.in/courses/115/103/115103123

4	https://nptel.ac.in/courses/112/105/112105051
5	https://nptel.ac.in/courses/112/105/112105050
6	https://nptel.ac.in/courses/115/107/115107116
7	Sensors and transducers https://nptel.ac.in/content/storage2/courses/112103174/pdf/mod2.pdf
8	https://www.vedantu.com/physics/anemometer
9	https://electricalacademia.com/renewable-energy/wind-speed-measurement-anemometer-types-working
10	https://circuitdigest.com/tutorial/solar-radiation-measurement-methods-using-pyrheliometer-and-pyranometer
11	https://solar-energy.technology/what-is-solar-energy/solar-radiation/measurement
11	https://www.hukseflux.com/applications/solar-energy-pv-system-performance-monitoring/how-to-measure-solar-radiation